

Process Logic, **Learned.**

A hybrid neural–symbolic pipeline that learns the hidden grammar of semiconductor manufacturing — and proves it.

Next-step

Completion

Anomaly detection

+ hidden 4th-family OOD

Team TBD

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A chip is a **150-step sentence** where order is physics.

- ▶ Every wafer runs an ordered route: **clean** → **oxidize** → **pattern** → **etch** → **implant** → ... → **test** → **ship** (~115–150 steps, 3 families).
- ▶ Order encodes real constraints: **no etch before a mask, no deposit on a dirty surface, no ship before sort-test** — 10 long-range rules.
- ▶ The judges' real question: does the model learn the process **logic**, or just memorize patterns?
- ▶ And it's graded on a **hidden 4th family** never seen in training.

Treat a recipe like a **sentence**. One step = one word.

TOKENIZER

A custom atomic-step tokenizer — each whole step is one token (vocab 210). No sub-word fragmentation; recipes read and write like language.

The Writer · DECODER

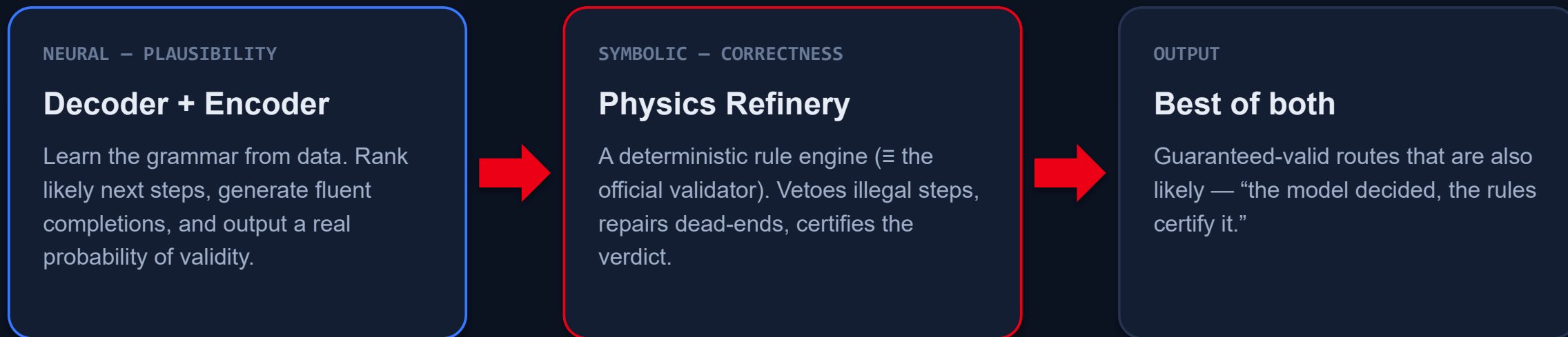
A from-scratch Llama-style causal Transformer that generates — next step & completion (Tasks 1–2). Attention does long-range lookups an LSTM can't.

The Judge · ENCODER

A from-scratch DeBERTa-v2 encoder that classifies validity + which rule broke (Task 3), trained with contrastive hard-negative twins.

Two specialized Transformers, trained from scratch — no pretrained model, full control of the vocabulary, interpretable representations.

The model proposes. **Physics disposes.**



Neural alone can be **confidently wrong**; rules alone can't say what's **likely**. The hybrid covers both blind spots.

One hybrid, wired **three ways**.

1 Next-step

Decoder's top-5 candidates, then the Refinery floats the legal ones to the top. Cannot hurt a good model; rescues a bad rank.

2 Completion

Decoder proposes; rule-vetoed beam search + repair (beam 5, branch 8). Every emitted suffix is provably valid.

3 Anomaly

Contrastive encoder learns WHY a route is invalid; rule engine gives the verdict, the model gives the graded score (real ROC-AUC).

Generative tasks use the writer; the discriminative task uses the judge — each with a guarantee the learned model can't make alone.

It works — and we can **prove it learned.**

Task	Headline	Also
1 · Next-step	Top-1 0.937 · Top-5 1.000	MRR 0.97 (hybrid)
2 · Completion	Block 0.94 · 100% rule-valid	token 0.71 · exact 0.28 · cat-tok 0.83
3 · Anomaly	F1 1.000 · rule-attr 0.972	encoder alone AUC 0.49 (honest)
Understanding	0.998 next-operation acc	learns function, not just names

learns **function**, not just names: predicts the right **operation 99.8% of the time** · 100% rule-valid · per-family T1: IGBT .955 / MOSFET .945 / IC .910

Honest provenance: real numbers from the organizer's official scorer on a held-out self-eval split, from the completed ProcSeq Leonardo run (base, 16k steps). The learned anomaly encoder alone is ~chance (AUC 0.49) — the physics hybrid carries Task 3. Organizer hidden-test and 4th-family OOD are pending.

Not just a score — evidence it learned the logic.

Learns the operation

0.998 next-OPERATION accuracy — it predicts the right kind of step (deposit / etch / clean), not just a memorized name. That's process logic, not pattern recall.

Always valid

100% of completions satisfy all 10 rules. Physics-vetoed beam search + repair means 0 illegal outputs — a guarantee a neural model alone can't give.

Honest on the hard task

The learned anomaly encoder alone is ~chance (AUC 0.49); the physics hybrid supplies the exact verdict (F1 1.0, rule-attr 0.97). We show both, not just the flattering one.

Every number is held-out and official-scorer; the organizer hidden test and 4th-family OOD are marked pending — we report what's measured.

Clear next steps, **already scoped.**

- ▶ **Scale the curve** — two larger model sizes / more steps to chart accuracy vs. compute.
 - ▶ **Ontology input channel** — feed each step's physical category into the encoder so it reads an unseen 4th family.
 - ▶ **The honest OOD curve** — run the novel-vocabulary stress test on the trained models (rename fraction $0 \rightarrow 1$).
 - ▶ **Calibrate + per-family** — calibrate the hybrid anomaly score and add a per-family results breakdown.
- ProcSeq — process logic, learned, guaranteed, and honestly benchmarked. **Thank you.**